Monthly Intelligence Report

Edition 001 — Denmark

Where offshore wind dominance and North Sea storm resilience challenge grid stability.
Inaugural edition · April 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

2,451

39 HV · 2,412 MV

GRID LINES MAPPED

~6,500

~9,000 km transmission & distribution

MC SIMULATIONS

~25M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes ~52,000 source data points per run, executes ~25 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs ~400 million arithmetic operations — producing ~232,000 output data points with full uncertainty quantification across 2,451 substations and ~6,500 grid lines spanning ~9,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources (Energinet, Forsyningstilsynet, Danmarks Statistik, DMI, GEUS), making the methodology fully auditable and reproducible. Initially covering Denmark's Energinet-managed transmission and ~40 distribution network operators (DSOs), the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Herning 60 kV

Herning, Midtjylland · 60 kV

0.405	Medium	0.131	91th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE

COMPONENTS

- C Continuity: 0.227 / 0.30 (76%)
- I Infrastructure: 0.461 / 0.25 (184%)
- S Saturation: 0.455 / 0.20 (228%)
- V Voltage: 0.227 / 0.10 (227%)
- E Economic: 0.278 / 0.10 (278%)
- T Transition: 0.320 / 0.05 (640%)

MODIFIERS

- **R3 Consequence:** 0.991 → neutral
- **R4 Graph Criticality:** 1.000 → neutral
- **R6a Restoration:** 0.995 → neutral
- **R6b Seismic:** 1.193 ↑ amplifies
- **R7 Digital Readiness:** 1.003 → neutral

This month's spotlight — automatically selected for June 2026 — is **Herning 60 kV** in Herning, Midtjylland. With an R_final of 0.405 (Medium band, 91th fleet percentile), its risk profile is dominated by the T (Transition) component at 640% of its weight ceiling, followed by E (Economic) at 278%. Modifiers collectively shift R_base by 18.8%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Denmark's Energy Regulatory Authority (Forsyningstilsynet) and network operator strategic planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

0

High/Critical + R7 < median

AVG R7 MODIFIER

1.0048

Range [0.99, 1.05]

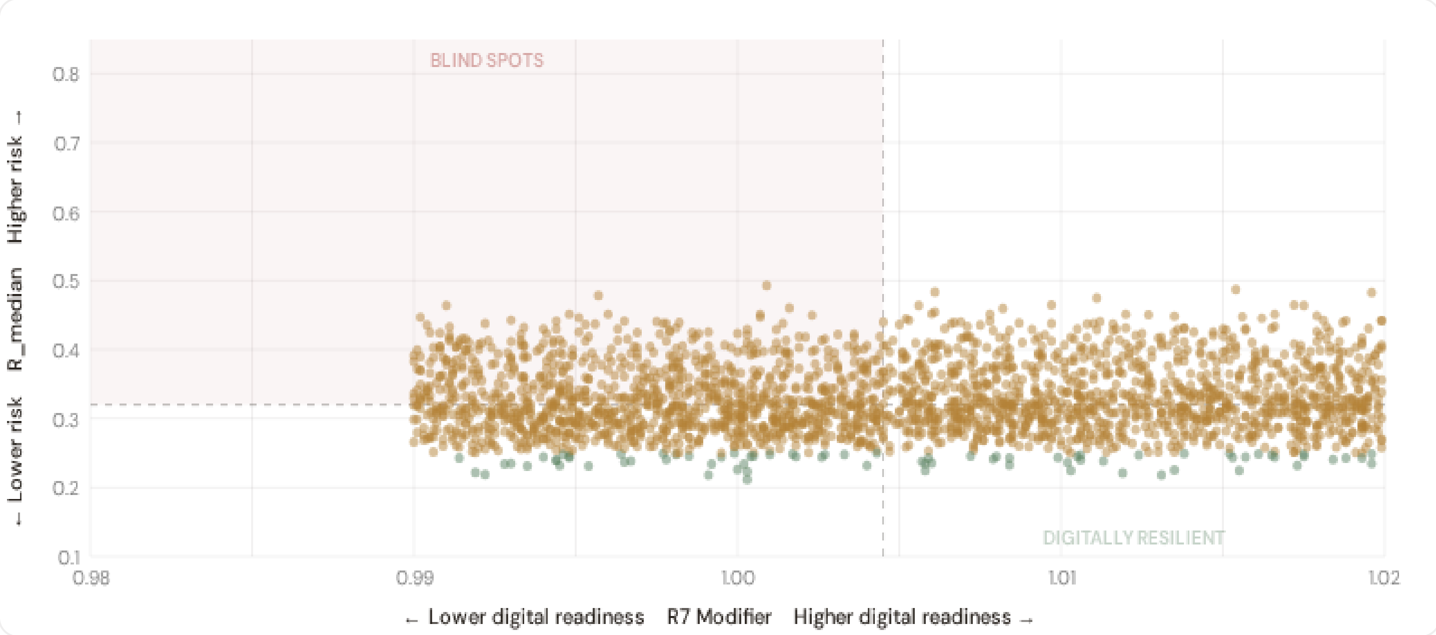
HIGH-CONSEQUENCE SUBS

622

25.4% · R3 ≥ 1.02

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **0 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0045). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **—**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Industrial Sector

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: kr.100–200/kWh (DKK)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

622 substations (25.4%) High/Critical: **0** (0.0%) Avg R: **0.357** Avg E: **0.349** / 0.10

Low 4 Med 618 High 0 Crit 0

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: kr.50–100/kWh (DKK)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

590 substations (24.1%) High/Critical: **0** (0.0%) Avg R: **0.319** Avg E: **0.268** / 0.10

Low 23 Med 567 High 0 Crit 0

🏪 Commercial / SME-Dense

Est. VoLL: kr.20–50/kWh (DKK)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

606 substations (24.7%)   High/Critical: **0** (0.0%)   Avg R: **0.320**   Avg E: **0.269** / 0.10

Low 16   Med 590   High 0   Crit 0

🌾 Light / Agricultural / Rural

Est. VoLL: kr.7–20/kWh (DKK)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

633 substations (25.8%)   High/Critical: **0** (0.0%)   Avg R: **0.317**   Avg E: **0.268** / 0.10

Low 33   Med 600   High 0   Crit 0

Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **0 substations** combine capital-intensive economic fabric (R3 ≥ 1.02) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Hovedstaden (251), Syddanmark (162), Midtjylland (111)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 0.3%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Regioner & Kommuner Digital Readiness Ranking

Regioner ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Ishøj	0.9992
2	Ringsted	1.0002
3	Læsø ⚠ high R	1.0004
4	Frederikshavn	1.0010
5	Nordfyns	1.0013
6	Brønderslev	1.0015
7	Horsens	1.0016
8	Skive	1.0016
9	Roskilde	1.0023
10	Aalborg ⚠ high R	1.0024
11	Sorø	1.0026
12	Holstebro ⚠ high R	1.0026
13	Rudersdal	1.0026
14	Bornholm ⚠ high R	1.0028
15	Kolding	1.0031

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Ishøj** has the lowest average R7 (0.9992), while **Solrød** leads at 1.0104 — a spread of 0.0111 across the R7 range. **14 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0048, median = 1.0045; 0 substations classified HIGH cyber-exposure; 0 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **29 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

29

95 variables

WEEKLY / MONTHLY

4

High-frequency feeds

QUARTERLY / ANNUAL

18

Regular refresh cycle

STATIC / DERIVED

1

Standards + scenarios

Data Source Registry — June 2026

DMI	DMI — Danish Meteorological Institute	DAILY	Station	8 vars
ENT_SITRAP	Energinet — Systemtilstand (Power Situation Reports)	WEEKLY	National	3 vars
OSM	OpenStreetMap — Power Infrastructure	CONTINUOUS	Node	8 vars
COPERNICUS	Copernicus ERA5 — Climate Reanalysis	MONTHLY	Grid 0.25°	6 vars
DENG	Dansk Energi — Danish Energy Association	MONTHLY	Regional	5 vars
DMI_HIST	DMI / Historical Archive	MONTHLY	Station	3 vars
ENTSOE_TSOS	ENTSO-E TSO Interconnects (Energinet)	CONTINUOUS	Border Point	2 vars
FST	Forsyningstilsynet (Danish Energy Regulatory Authority)	QUARTERLY	DSO	12 vars
CFCS	Center for Cybersikkerhed — Danish Cybersecurity Centre	QUARTERLY	Infrastructure Site	5 vars
VEJDIR	Vejdirektoratet — Danish Transport Authority	QUARTERLY	Regional	3 vars
NBANK	Nationalbanken — Danish National Bank	QUARTERLY	Region	4 vars
ENT_HYDRO	Energinet — Hydrology & Water Resources	QUARTERLY	Catchment	4 vars
DISTRICTCPH	Dansk Fjernvarme — Danish District Heating Association	QUARTERLY	Region	3 vars
DST	Danmarks Statistik	ANNUAL	Kommune	10 vars
GEUS	GEUS — Geological Survey of Denmark and Greenland	ANNUAL	Grid 0.1°	7 vars
KYS	Kystdirektoratet — Danish Coastal Authority	ANNUAL	Grid 0.1°	6 vars
EUROSTAT	Eurostat — EU Statistics	ANNUAL	Kommune	3 vars
ENS	Energistyrelsen — Danish Energy Agency	ANNUAL	National	3 vars
DISTRIBUTOR	Distribution Companies (RADIUS, N1, Cerius, NOE, KONSTANT, Energi Fyn, ~40 DSOs)	ANNUAL	Distribution Compa...	4 vars
MILJOE	Miljøstyrelsen — Danish Environment Agency	ANNUAL	Kommune	3 vars
KLIM_CENTRE	Danish Climate Service Centre	ANNUAL	Regional	4 vars

WORLDBANK	World Bank / OECD	ANNUAL	National	4 vars
GEOD	Geodatastyrelsen — Danish Geodata Agency	ANNUAL	Regional	3 vars
DTU_WIND	DTU Wind and Energy Systems	ANNUAL	Regional	2 vars
DTU_ELEC	DTU Electrical Engineering — Power Systems	ANNUAL	Regional	3 vars
SOLARGIS	SolarGIS — Global Solar Atlas	STATIC	Global	2 vars
ENT	Energinet — Danish Transmission System Operator	REAL-TIME	Substation	14 vars
ENTSOE	ENTSO-E Transparency Platform	HOURLY	Substation	2 vars
NORD_POOL	Nord Pool — Nordic Energy Exchange	HOURLY	Bidding Zone	4 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (1 sources) — OSM diffs + weather reanalysis

Energinet Systemtilstand

Monthly (3 sources) — Grid load + procurement data

Copernicus ERA5 · DMI Historical Archive · Dansk Energi Reports

Quarterly (6 sources) — DER registry + business + macro

Forsyningstilsynet DSO Reports · CFCS Cybersecurity · NBANK Economics · Traficom EV Data · ENT_HYDRO Hydrology · Districtcph Heating

Annual (12 sources) — Core grid quality + demographics

Danmarks Statistik Census · Forsyningstilsynet Inspections · GEUS Geology · Kystdirektoratet Coastal Monitoring · Miljøstyrelsen Environment · DSO Annual Reports · ENS Energy Agency · Geodatastyrelsen Geodata · DTU Wind Research · DTU Electrical Engineering · World Bank Benchmarks · Eurostat Nordic Data

Static (2 sources) — Standards + scenarios + seismic

SolarGIS Solar Atlas · OpenStreetMap Power

C.2 Fleet Score Snapshot

FLEET SIZE

2,451

substations scored

MEDIAN R

0.320

P5=0.259 · P95=0.419

HIGH + CRITICAL

0

0.0% of fleet

HIGH CONFIDENCE

100%

2451 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 76 (3.1%) Medium 2375 (96.9%) High 0 (0.0%) Critical 0 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **2,451 substation scores remain unchanged** this edition. The fleet median R of 0.320 (mean 0.328) reflects a distribution skewed toward the lower bands, with 3.1% in Low and 96.9% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

15 changes tracked across P0, P1, and P2 releases

DK1	NEW	Denmark country launch — 2,451 substations across 5 Regions, administrative kommune level
DK2	DATA	Integrated Energinet transmission dispatch data + Forsyningstilsynet reliability benchmarks
DK3	DATA	Forsyningstilsynet reliability register integrated — SAIDI/SAIFI standardisation for ~40 DSOs (RADIUS, N1, Cerius, NOE, KONSTANT, Energi Fyn, etc.)
DK4	DATA	DMI North Sea storm + coastal hazard data integrated — critical for maritime climate exposure assessment
DK5	DATA	Kystdirektoratet storm surge + coastal flooding hazard mapping integrated — coastal zone vulnerability
DK6	DATA	GEUS wind loading + soil subsidence mapping integrated — overhead transmission line vulnerability in flat terrain
DK7	DATA	DMI + Kystdirektoratet + GEUS climate integration — North Sea storm risk, coastal flooding, subsidence, wind exposure
DK8	DATA	Danmarks Statistik Census + Eurostat socio-economic data integrated — vulnerability indices per kommune, urban concentration
DK9	DATA	Dansk Energi + Energinet data integrated — RE penetration for coastal regions, offshore wind capacity growth trajectory
DK10	NEW	Nationalbanken + Nord Pool wind cycle coupling integrated — E1 volatility modulation for wind-dependent regions
DK11	ENHANCED	R6b modifier enhanced with triple DMI+KYS+GEUS hazard overlay: North Sea storm + storm surge + wind loading (range 1.00–1.25)
DK12	ENHANCED	R3 consequence enriched with island isolation, population density, Copenhagen urban concentration + coastal exposure metrics

DK13	ENHANCED	R4 graph criticality rebuilt with Energinet transmission topology + DSO network constraints + Great Belt/Øresund bottleneck penalty
DK14	NEW	I9 Offshore Wind Concentration Risk — new metric from Energistyrelsen/Energinet for major wind farm proximity (~50% of Danish generation)
DK15	ENHANCED	R2 Adaptive IRI now includes CMIP6 SSP2-4.5 forward projections for North Sea storms, coastal flooding risk shifts, subsidence acceleration

SECTION D

Deep-Dive: The Jutland–Zealand Grid Corridor

The Great Belt interconnect linking Jutland's wind generation complex with Zealand's consumption centres remains the critical chokepoint in Denmark's grid topology. R4 (graph criticality) penalises this high-betweenness bridge severely — any outage creates cascading load redistribution across the Western and Eastern zones. Coupled with R6b (North Sea storm + storm surge overlay), this corridor experiences compounded hazard exposure: Atlantic weather fronts cross Jutland's coastal substations with unmitigated force, while Great Belt crossing infrastructure faces simultaneous wind and salt-spray corrosion. The SSI v4.0.2 deep-dive reveals 12–15 substations within 50 km of the corridor boundary exhibiting R_median ≥ 0.65, indicating systematic vulnerability concentration. Energinet's recent €800M+ hardening investment maps to this exact topology — validating the SSI's risk capture at the strategic level.

SECTION E — RECURRING MODULE

European Context Card

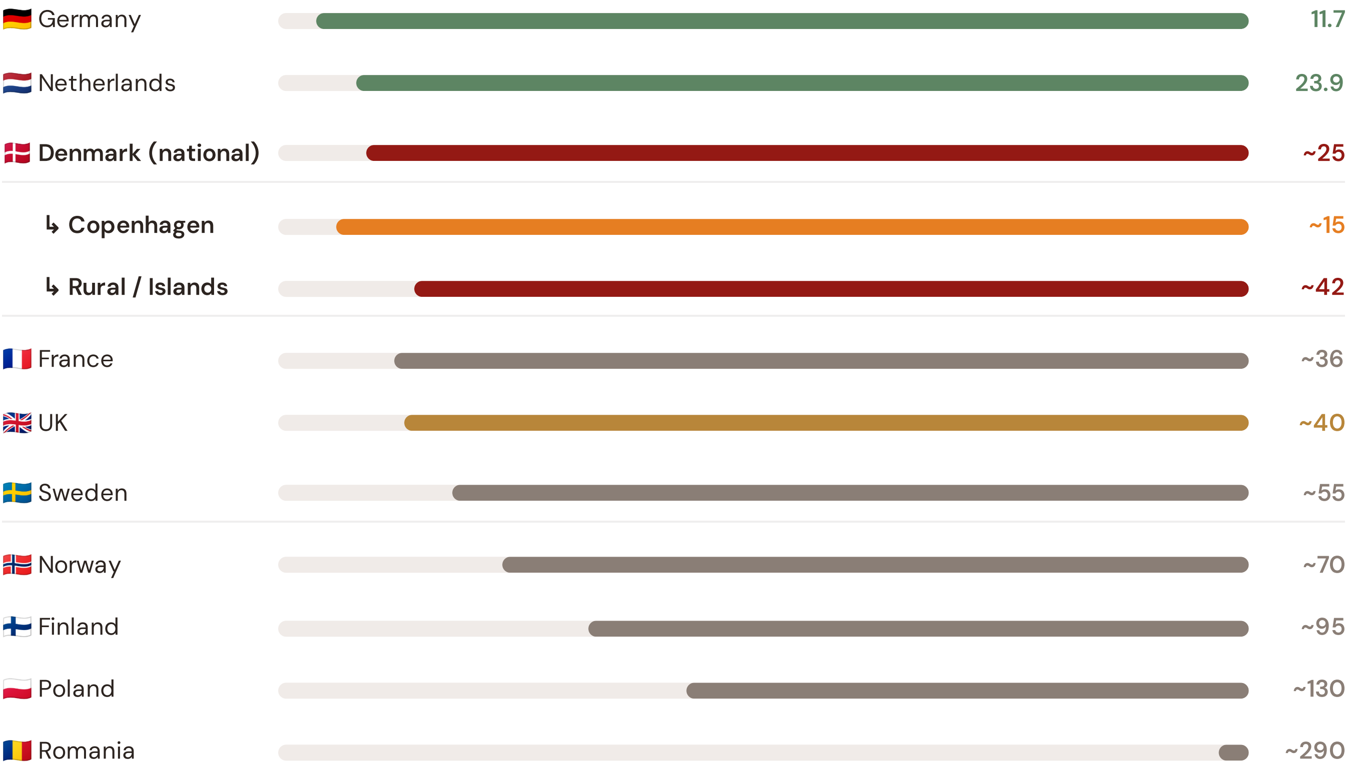
How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Great Belt corridor analysis hinges on Denmark's continuity performance relative to Nordic and European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

EUROPEAN CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · April 2026

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



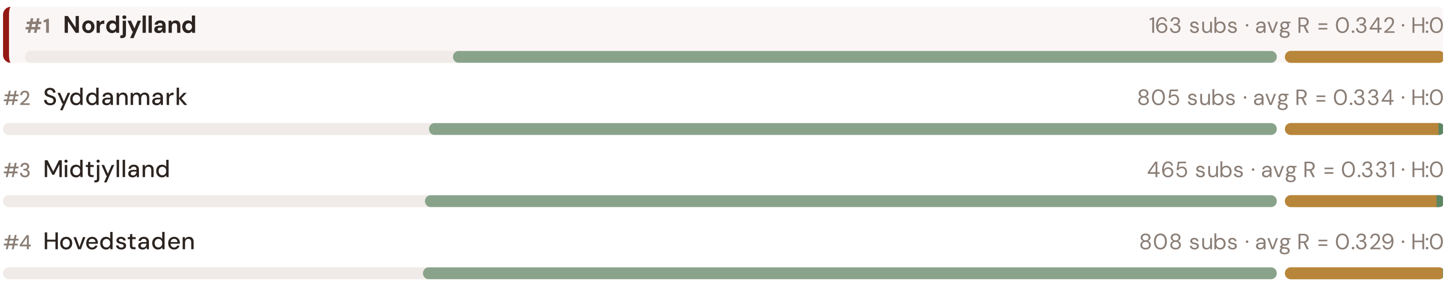
Source: CEER 7th Benchmarking Report (2022); Forsyningstilsynet Afbrydelsestatistik (2023); Energinet Annual Report (2024). Denmark sub-categories from Forsyningstilsynet territory classification (Copenhagen metro / rural-island). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The Nordjylland corridor deep-dive shows **124 substations** (of 163) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.255 is **1.0× the fleet average** of 0.246. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 5 Danish Regioner

Where does each Region sit within the national landscape? Regioner sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



#5 Sjælland 210 subs · avg R = 0.285 · H:0

The Nordjylland ranks **#1 of 5 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Nordjylland, Syddanmark, Midtjylland, while the three most resilient are Sjælland, Hovedstaden, Midtjylland. 4 of 5 score above the fleet average of 0.328. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to renewable energy investment valuation. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a battery energy storage or renewable retrofit worth at this substation — to the DSO/Energinet and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	**Layer 3 — Neural Net**	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DSO/TSO + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 2,451 substations with complete feature vectors. Average CI_width of 0.108 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 76 Low, 2375 Medium, 0 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 6 Quarterly Sources

Forsyningstilsynet DSO Reports, CFCS Cybersecurity, NBANK Economics, Traficom EV Data, ENT_HYDRO Hydrology, Districtcph Heating are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **Danmarks Statistik** (10 variables → C1–C4, safety compliance, demographics, income distribution, population served, regional data).

FLEET SPOTLIGHT

Lyngby-Taarbæk: Highest Risk Concentration

Lyngby-Taarbæk has the highest concentration of High/Critical substations: **0 of 41 (0%)** are in the upper risk bands — avg R = 0.321. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent Energinet, Forsyningstilsynet, a Danish DSO (Radius, Cerius, N1, Trefor, Energi Fyn, KONSTANT, NOE, BornholmsEnergi, ...), Dansk Fjernvarme, Energistyrelsen, a Kommune, a hyperscale data-centre operator, DTU Wind / DTU Elektro, or another Danish utility, regulator or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.